

A Federated Expert-Fusion Framework for Cross-Bank Fraud Detection

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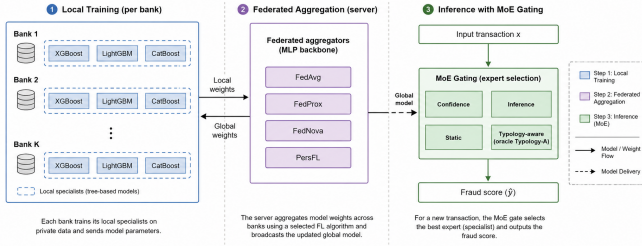


Fig. 1. Three-phase expert-fusion pipeline: per-bank tree specialists, federated MLP backbone aggregation across four banks, and a fusion gate at inference (Static, Performance, Confidence, or oracle Typology-Aware†). Each bank’s gate fuses its own three trees with the four shared backbones; no tree crosses a bank boundary.

Abstract—Banks cannot pool transaction data for fraud detection, so we measure what federated learning (FL) actually delivers across simulated bank silos. We benchmark four FL algorithms, three bank-local tree experts and four expert-fusion gates on three public fraud datasets under Dirichlet non-IID splits ($\alpha \in \{0.05, 0.10, 0.50\}$) and five seeds: 516 runs, every partition reproducible from (seed, dataset, α) alone. Three findings stand out. First, the deployable fusion gates hold the best mean rank of any family and beat the FL family average on all three datasets (Wilcoxon $p \leq 0.030$), yet never beat the per-cell best backbone ($p \geq 0.76$): fusion buys robustness against selecting a poor backbone, not raw accuracy. Second, FL collapse at extreme skew is a partition sensitivity rather than an algorithm ranking, and personalised FL never records $F1 = 0$. Third, a cost-aware triage layer turns rankings into decisions: at recall 80 it auto-clears 74 to 99.8% of transactions, defers 4 to 5% to human review, and its expected cost moves under 4% while the assumed cost ratio varies fiftyfold.

Index Terms—federated learning, fraud detection, expert fusion, mixture of experts, non-IID data, selective prediction

I. INTRODUCTION

A. Background

Money laundering runs on fragmentation. Worth an estimated US\$2 to 3.1 trillion a year [1], the typologies that hide it (layering, smurfing, mule networks) deliberately split funds across 4 to 7 banks [2], [3], so one bank sees one slice and single-institution monitoring catches roughly 5%. No bank holds enough of any one typology to train a detector

that generalises. The direct fix, pooling the data, is illegal: GDPR, the Australian Privacy Principles, APRA Prudential Standard CPS 234 [4] and competitive sensitivity all rule it out. The question therefore shifts from “can we share data?” to “can we share models?”. That is the regime McMahan et al. [5] built federated learning (FL) for: clients trade model weights, never raw records.

FL does not eliminate the difficulty; three pressures stack. First, imbalance: fraud prevalence runs from 0.10% (SAML-D, IBM HI-Small) to 0.172% (ULB), low enough that AUROC saturates and AUPRC becomes the only metric worth reading. Second, the non-IID split: banks specialise, so each client sees its own mix of typologies, producing client drift [6], [7]. Third, fraud labels in SAML-D span 17 to 28 typologies [2], so any global model is secretly a mixture model. At extreme non-IID the three compound and the global model can cave in.

B. Related Work

We screened 510 papers under PRISMA 2020 [8] and kept 42 with end-to-end FL or MoE results on financial data. They fall into three groups, each leaving a gap this paper addresses.

The first covers aggregation rules. FedAvg [5] computes a data-weighted average and remains the default baseline, reported to degrade sharply under non-IID; FedProx [9] adds a proximal term; FedNova [10] normalises for unequal local step counts. Our sweep tests how much of that reported fragility survives a multi-seed protocol.

The second personalises. PersFL/Per-FedAvg [11] keeps client-specific weights alongside shared ones; Ditto [12] splits the two components. The Dirichlet partition of Hsu et al. [13] is the standard non-IID simulator, and we adopt it. In our grid PersFL is the only algorithm that never records $F1 = 0$, at the price of a single unified global model.

The third combines experts. Adaptive mixtures [14] and sparsely gated MoE [15] were extended by expert-choice routing [16]; federated variants followed, with FedMix [30], Marfoq et al. modelling each client as a mixture of shared distributions [31], and FedJETs routing to specialist subsets [32]. pFL-Bench [33] gives personalisation the multi-seed benchmark treatment we attempt for fraud. Our fusion layer is deliberately simpler than these learned routers: per-bank scalar weights set on validation folds, no input-conditional routing, hence “expert fusion”. The strongest standalone tabular models remain gradient-boosted trees (XGBoost [17], LightGBM [18], CatBoost [19]), our local experts; federated boosting [34] targets vertical partitions, and direct FL-for-card-fraud work [35] is single-dataset, single-seed. A separate selective-prediction literature, from Chow’s rejection

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rule [28] to conformal prediction [29], asks when a classifier should defer to a human; fraud work rarely connects it to FL.

Similar limitations recur across all three groups: single seeds that hide non-IID variance, single datasets, AUPRC-only reporting, and oracle gates mixed in with deployable ones. To our knowledge no prior work evaluates multiple fraud datasets, FL methods, fusion strategies, seeds, cost-aware metrics and a deferral policy in one framework.

C. Research Questions and Contributions

Five research questions follow. RQ1: how does each FL algorithm degrade as the Dirichlet concentration α falls from 0.50 to 0.05? RQ2: can an expert-fusion layer bridge the divide between federated and centralised learning? RQ3: which aggregation rule should a practitioner pick for a given non-IID regime? RQ4: what is the utility cost of gradient noise and sparsification? RQ5: can a cost-aware triage layer convert performance rankings into an operational decision policy?

The contribution is a benchmark and a decision protocol rather than a new learning algorithm. First, a matched-protocol non-IID benchmark: eleven strategies (four FL algorithms, three tree experts, four fusion gates) $\times$ three dataset families $\times$ three α settings $\times$ five seeds gives 495 main-sweep runs, and a 60-round ULB noise-robustness grid brings the total to 516, with every partition seed derived only from (seed, dataset, α). Second, a two-convention statistical reading of the fusion layer: the gates beat the FL family mean on all three datasets and hold the best overall Friedman rank, yet never beat the per-cell best backbone, and their weights sit close to uniform, so the layer is a weighted ensemble rather than a router. Third, an algorithm-survival map indexed by non-IID regime and reported as seed frequencies rather than verdicts. Fourth, a triage layer (per-bank conformal thresholds, a Chow-style defer band, a capacity cap) that auto-clears 74 to 99.8% of transactions at recall 80 with a 4 to 5% defer rate.

II. METHODS

A. System Overview

The pipeline has three phases (Fig. 1). Phase 1 trains each bank's specialist classifiers on its private slice. Phase 2 federates a shared MLP backbone across banks. At inference, Phase 3 sends each transaction through a fusion gate blending local specialists with the federated backbones. Phase 2 reuses Phase 1's partition, and the Phase 3 gates are tuned on the per-bank validation folds the earlier phases leave behind. Seven experts sit under each bank's gate: the four shared FL-MLP backbones (PersFL contributes that bank's personalised copy) and the bank's own three trees, which never leave it.

B. Data Pipeline

Three public benchmark families stress-test the pipeline across scales and feature regimes. ULB Credit Card [20] contains 284,807 transactions at 0.172% fraud with 29 PCA features. SAML-D [2] is larger: 9.5 million transactions at

0.10% fraud, with 10 engineered features carrying typological structure. IBM AML [21] is used at HI-Small scale, roughly 5.1 million transactions at 0.10% fraud, whose held-out window contains 761,753 transactions with 1,561 frauds (0.21%). Every dataset passes through a $\log_{1p}$ transform on Amount and removal of the raw Time column, with StandardScaler fitted on each bank's training fold only.

Federated partitioning simulates $N = 4$ banks by drawing fraud indices from a Dirichlet prior and distributing legitimate indices uniformly. Given the positive index set I^+ and concentration α :

$$p \sim \text{Dir}(\alpha \cdot \mathbf{1}_N), \quad n_i^+ = p_i \cdot |I^+|, \quad n_i^- = |I^-|/N \quad (1)$$

The α sweep covers $\{0.05, 0.10, 0.50\}$. The partition RNG is seeded by a hash of (seed, dataset, α) alone, so any cell reproduces by itself. Per-bank splits are temporal 70/15/15. Class imbalance is handled with per-bank SMOTE [22] at $k_neighbors = \min(3, n_fraud - 1)$; for all tree models $scale_pos_weight$ is n_neg/n_pos .

C. Phase 1: Per-Bank Local Specialists

Each bank trains three gradient-boosted classifiers locally: XGBoost ($n_estimators = 200$, $max_depth = 6$, $eval_metric = aucpr$) [17], LightGBM ($n_estimators = 200$, $max_depth = 6$) [18] and CatBoost (iterations = 200, depth = 6, $eval_metric = PRAUC$) [19]. Tree ensembles remain dominant for tabular fraud, and each bank's local distribution stays well-conditioned whatever happens elsewhere in the federation.

D. Phase 2: Federated MLP Aggregation

The backbone is a scikit-learn MLPClassifier with hidden layers (128, 64, 32), ReLU activations, Adam ($lr = 10^{-3}$), batch size 128 and log-loss, trained for 20 rounds. Each round every bank continues from the current global weights, so an initialised network is never reinitialised. FedAvg [5] aggregates client weights by sample count (2). FedProx [9] adds a proximal regulariser at $\mu = 0.01$ (3); sklearn exposes no per-step loss hook, so the term is applied per epoch as a documented pull $\min(\eta\mu S, 0.5)$ toward the global weights ($S = \text{steps per epoch}$). FedNova [10] normalises each client's update by its local step count $\tau_i = 5 \lceil n_i/128 \rceil$ before aggregation (4). PersFL [11] blends bank i 's weights 50/50 with the global model after each local fit, and those personalised copies are what Phase 3 evaluates.

$$w^{(t+1)} = \sum_i (n_i/N_{\text{tot}}) \cdot w_i^{(t)} \quad (2)$$

$$L_i(w) = \text{BCE}(w; D_i) + (\mu/2) \|w - w^{(t)}\|^2, \quad \mu = 0.01 \quad (3)$$

$$w^{(t+1)} = w^{(t)} + \tau_{\text{eff}} \sum_i [n_i \tau_i / \sum_j n_j \tau_j] d_i \quad (4)$$

E. Phase 3: Expert-Fusion Gating

Write $p_e(x)$ for expert e 's probability on transaction x . Each bank's gate fuses its seven experts as $\hat{y}(x) = \sum_e g_e \cdot p_e(x)$, and the four variants differ only in how they set g_e . The weights are per-bank scalars computed once on the validation fold; nothing is input-conditional and no

router is learned. Static is the baseline, $g_e = 1/E$. Performance weights by validation F2, $g_e = F2_e^{\text{val}} / \sum_{e'} F2_{e'}^{\text{val}}$. Confidence rewards experts whose scores sit far from the decision boundary, defined in (5), and never looks at a label. Typology-Aware[†] routes on the ground-truth fraud type t , $g_e(x|t) = F1_e^{\text{val}}(t) / \sum_{e'} F1_{e'}^{\text{val}}(t)$, which is why it cannot ship: the label is absent at inference, so we carry it as an upper bound only, tagged [†] wherever it appears.

$$g_e = c_e / \sum_{e'} c_{e'}, \quad c_e = \mathbb{E}_x[|p_e(x) - 0.5| \times 2] \quad (5)$$

F. Gradient-Noise and Sparsification Robustness

This ablation predates the canonical sweep and runs on a separate 60-round ULB implementation with a four-layer PyTorch backbone, five banks and $\alpha = 0.5$. Two mechanisms are measured, both as robustness experiments. *Gaussian gradient noise*: gradients are clipped each step to norm $C = 1.0$, then noise is added with $\sigma = zC/n_b$ at $z = 1.1$, the mechanism (though not the accounting) of DP-SGD [23], [24]. There is no privacy accountant and no differential-privacy claim; an ε of 4.0 quoted in earlier drafts is withdrawn. *Top-K sparsification* [25]: only the top 1% of gradient entries by magnitude are transmitted each round, cutting communication by roughly two orders of magnitude.

G. Evaluation Protocol

AUPRC is the primary metric, since AUROC turns overly optimistic when the positive class sits near 0.1 to 0.2%. Secondary metrics are final-round F1 at the actual operating point rather than the best across rounds, F2, Matthews correlation coefficient, and per-bank worst-case F1. All main results are 5-seed mean $\pm$ std.

For statistical testing we use a Friedman χ^2 test over the 45 seed-condition blocks [27], followed by Wilcoxon signed-rank tests under two conventions, family means (does a gate beat the average backbone?) and per-cell bests (does the best gate beat the best backbone?), with Cliff’s delta effect sizes [26]. With five seeds a per- α two-sided Wilcoxon cannot go below $p = 0.0625$, so the pooled 15-block tests carry the p-values, with a resolution floor of $p = 6.1 \times 10^{-5}$ at roughly 40 single-GPU hours. Tables report raw p-values; every significance call survives a Holm–Bonferroni adjustment within its comparison family. A cost-aware layer varies the relative cost of false negatives: expected loss is $L(\rho) = \text{FN} \cdot \rho + \text{FP}$, across $\rho \in \{10, \dots, 5000\}$. On hyperparameters our policy is symmetry over search: both expert families run library-typical settings at matched budgets with no per-dataset tuning.

H. Reproducibility

All experiments ran across five seeds (42, 0, 1, 2, 3) on single NVIDIA Tesla P100 16 GB GPUs on Kaggle’s standard image (Python 3.12). Because the partition seed depends only on (seed, dataset, α), the grid splits across sessions and machines onto identical cells; the repository contains the sweep scripts, the triage package and complete per-cell outputs. Two forensic findings shaped this section: a

TABLE I
CONSOLIDATED HYPERPARAMETERS (S = SWEEP, G = PRIVACY GRID)

Param	Value	Param	Value
Banks N	4 s / 5 g	n_est. / depth	200 / 6 s
Rounds $\times$ epochs	20 $\times$ 5 s / 60 $\times$ 3 g	SMOTE k	$\min(3, n_{\text{fraud}} - 1)$
MLP hidden	(128, 64, 32) s	Noise C, z	1.0, 1.1 g
Partition seed	hash(seed, data, α)	Top-K	0.01 g
PersFL blend	0.5 local / 0.5 global	Recall target	0.80
FedNova τ_i	5 $\lceil n_i / 128 \rceil$ s	Defer cap	5%
Batch / LR	128 / 1×10^{-3}	FedProx μ	0.01
Dirichlet α	{.05, .10, .50} s / 0.4 g	Seeds	{42, 0, 1, 2, 3}

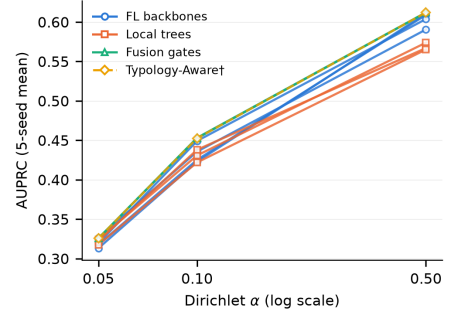


Fig. 2. ULB AUPRC vs Dirichlet α , 5-seed means, coloured by family. Degradation is monotonic ($\rho = +0.943$) and family ordering is stable.

reinitialising FL loop, whose byte-identical rows fabricated the retracted 0.699 ULB plateau and the FedProx tenfold claim, and RNG tracing that showed our earliest result set came from unmerged code. The PyTorch privacy grid keeps one caveat: its global seed was not fixed, so its numbers carry floating-point-level noise. Table I lists the hyperparameters.

III. EXPERIMENTS AND RESULTS

Eleven strategies, three dataset families, three Dirichlet settings and five seeds give 495 main-sweep runs, plus 21 populated cells of a 60-round ULB noise-robustness grid, 516 in total. Every headline below is a 5-seed mean $\pm$ std unless stated otherwise.

A. Heterogeneity Sweep (RQ1)

Take AUPRC for the eleven strategies’ seed means on a dataset and correlate with α . On ULB that correlation is strong and positive: Spearman $\rho = +0.943$, $p = 2 \times 10^{-16}$, $n = 33$. The fall-off is gentle (FedAvg means 0.609 at $\alpha = 0.50$, 0.425 at 0.10, 0.318 at 0.05), and Fig. 2 shows it holding for every backbone and gate, the oracle included.

The AML datasets behave differently, and differently from each other. On SAML-D, $\rho = +0.772$ ($p = 1.5 \times 10^{-7}$), but the mean hides the mechanism: at $\alpha = 0.05$ four of five seeds send every standard backbone to AUPRC 0.002 to 0.004 together, while the fifth leaves them all intact around 0.09. Survival at extreme skew is decided by the partition draw; FedAvg and FedProx sit within a hair of each other in every cell. On IBM HI-Small the correlation disappears altogether ($\rho = +0.150$, $p = 0.40$): performance is nearly flat in α (FedAvg 0.033 / 0.027 / 0.039), so what limits IBM is feature signal. Non-IID degradation is thus three curves: graceful on PCA features, a partition-induced instability on SAML-D, mostly absent on IBM.

TABLE II
 $\alpha = 0.5$ CROSS-DATASET AUPRC, 5-SEED MEAN $\pm$ STD

Strategy	ULB	IBM HI-S	SAML-D
FedAvg	0.609 $\pm$ 0.151	0.039 $\pm$ 0.005	0.075 $\pm$ 0.027
FedProx	0.604 $\pm$ 0.148	0.037 $\pm$ 0.009	0.070 $\pm$ 0.026
FedNova	0.611 $\pm$ 0.151	0.043 $\pm$ 0.006	0.071 $\pm$ 0.026
PersFL	0.590 $\pm$ 0.122	0.033 $\pm$ 0.008	0.073 $\pm$ 0.021
XGBoost	0.574 $\pm$ 0.151	0.043 $\pm$ 0.002	0.080 $\pm$ 0.013
LightGBM	0.567 $\pm$ 0.153	0.051 $\pm$ 0.001	0.079 $\pm$ 0.013
CatBoost	0.565 $\pm$ 0.157	0.044 $\pm$ 0.002	0.077 $\pm$ 0.007
Fusion-Static	0.612 $\pm$ 0.151	0.044 $\pm$ 0.003	0.076 $\pm$ 0.027
Fusion-Perf	0.612 $\pm$ 0.151	0.044 $\pm$ 0.003	0.077 $\pm$ 0.027
Fusion-Conf	0.612 $\pm$ 0.151	0.044 $\pm$ 0.003	0.076 $\pm$ 0.027
Fusion-TypAw†	0.612 $\pm$ 0.151	0.051 $\pm$ 0.005	0.083 $\pm$ 0.020

† Oracle gate; upper bound only, never deployed.

B. Expert Fusion Under Two Conventions (RQ2)

Table II is the $\alpha = 0.5$ cross-dataset cell. On ULB the gates (0.612 $\pm$ 0.151) top the deployable table, a nose above the best backbone (FedNova 0.611) and clearly above the trees (0.565 to 0.574). On SAML-D and IBM HI-Small the trees lead (0.079, 0.051) with the gates just behind, well clear of FL. The oracle escapes the pack only on the AML datasets (0.051 IBM, 0.083 SAML-D), bounding what a better typology detector could add. On ULB it cannot escape by construction: with one fraud typology its weights fall back to the performance gate’s for 99.8% of transactions, so the routing headroom is exactly zero.

Whether the gate “wins” depends on the question, so Table III asks both. Against the FL *family mean* the gates win on every dataset (pooled 15-block Wilcoxon $p = 0.030 / 6 \times 10^{-5} / 0.012$); against the *per-cell best* backbone they are a coin flip everywhere ($p \geq 0.76$). Both are true at once: the gate is a hedge. Practitioners cannot know in advance which backbone a partition will favour, and the near-uniform blend lands at or above the field’s centre of mass. The measured weights agree: the top expert’s mean weight stays between 0.143 (uniform for seven experts) and 0.29, and the nominal top expert flips with the seed. As an ablation of the weighting function the verdict is that it barely matters: static, F2-weighted and confidence weighting land within 0.001 AUPRC of one another everywhere. The reason is mechanical. Linearly normalising validation scores that all sit in a narrow band gives near-uniform weights by construction, and on 99.8%-negative data the confidence signal mostly records sigmoid saturation. Expert rankings also correlate at 0.37 to 0.93, and AUPRC is rank-based, so perturbations this small have nothing to move. Nor is this an implementation artefact: replayed on the captured probabilities, sharpened weights change nothing and rank-normalising before blending *costs* 0.015 AUPRC on ULB and SAML-D.

Against the centralised pooled baseline the ceiling holds everywhere: XGBoost reaches 0.764 on ULB, CatBoost 0.071 on IBM and 0.089 on SAML-D, leaving the best deployable method (Fusion-Static 0.612, Fusion-Perf 0.044 and 0.077) 0.152, 0.027 and 0.012 short. These baselines are single pooled runs on a fixed temporal split; XGBoost was byte-identical across three sessions and CatBoost moved by ≤ 0.001 , while centralised LightGBM is degenerate and

TABLE III
DEPLOYABLE FUSION GATES VS OTHER FAMILIES, POOLED 15-BLOCK WILCOXON (P / CLIFF’S δ)

Comparison	ULB	IBM HI-S	SAML-D
vs FL, family mean	.030 / +.10	6e-5 / +.54	.012 / +.17
vs FL, per-cell best	.80 / -.00	.76 / +.12	.85 / +.04
vs trees, family mean	1.2e-4 / +.18	.083 / -.56	.055 / -.23
vs trees, per-cell best	.002 / +.16	.011 / -.72	.018 / -.25

Bold: $p < 0.05$, all surviving Holm–Bonferroni. Positive δ favours the gates; on IBM and SAML-D the best-vs-best results favour the *trees*.

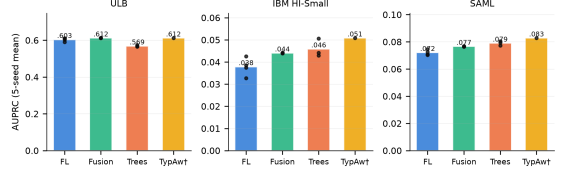


Fig. 3. Family-level AUPRC at $\alpha = 0.5$; bars are family means, dots individual strategies. Note the per-panel scales.

excluded pending a fix. One earlier draft had a seed-42 cell beating the baseline by +0.089; the audited pipeline puts that cell 0.181 *below* it. Over all 45 blocks the Friedman test ranks the deployable families gates 4.99, trees 5.61, FL 7.77 ($\chi^2 = 110.2$, $p = 5 \times 10^{-19}$; the oracle sits at 3.11), and Fusion-Perf is the best single deployable strategy at mean rank 4.48. RQ2 therefore gets a two-part answer: fusion is the best deployable family on average, and the gap to pooled raw data does not close.

C. FL Algorithm Survival (RQ3)

Table IV counts, for each backbone, the seeds whose final F1 lands at exactly zero, turning operating-point collapse into a frequency. Three patterns emerge that the deterministic verdicts of earlier drafts had flattened. First, collapse at $\alpha = 0.05$ is a partition-dependent variability, and a largely strategy-blind one: on SAML-D, seeds 0, 1, 3 and 42 take every standard backbone down together while seed 2 spares them all, and on IBM exactly one seed zeroes FedAvg, FedProx and FedNova at once. Second, FedNova owns the most fragile operating point: no other backbone hits F1 = 0 in four of five SAML-D seeds at $\alpha = 0.05$, and at $\alpha = 0.10$ it is the only backbone to record a zero at all. Third, PersFL never records F1 = 0 anywhere, posting 2 to 10 $\times$ the AUPRC of the other backbones inside collapsed partitions, though two SAML-D seeds still leave its AUPRC under 0.005.

Cost re-ranking sharpens the deployment picture. Across the 81 (dataset, α , ρ) cells, the method with the best AUPRC is also the cost-best deployable method in only 10; the gates take 45 cells, concentrated at $\rho \leq 50$ (17 of 27), and PersFL takes 23, concentrated at $\rho \geq 1000$ (15 of 27). Review-cheap regimes favour a gate, miss-expensive regimes favour personalisation.

D. Gradient-Noise and Sparsification Robustness (RQ4)

Table V reports final-round AUC from the 60-round ULB grid. This grid predates the canonical sweep (PyTorch, single seed), so we read it for deltas rather than absolute levels. The noise result is the most notable: relative to clean runs,

TABLE IV
OPERATING-POINT COLLAPSE FREQUENCY: SEEDS (OF 5) WITH $F1 = 0$.
U/S/I = ULB/SAML-D/IBM HI-SMALL

Algo	U.5	U.1	U.05	S.5	S.1	S.05	I.5	I.1	I.05
FedAvg	0	0	0	0	0	1	0	0	1
FedProx	0	0	0	0	0	1	0	0	1
FedNova	0	0	0	0	1	4	0	0	1
PersFL	0	0	0	0	0	0	0	0	0

TABLE V
ULB 60-ROUND FINAL-ROUND AUC UNDER GRADIENT NOISE
(SINGLE SEED, SEED 42)

	MLP	MLP	MLP	LR	LR	LR
FL algo	Clean	Noise	Spars	Clean	Noise	Spars
FedAvg	0.966	0.973	0.945	0.960	0.958	0.955
FedProx	0.974	0.977	0.957	0.960	0.963	0.957
FedNova	0.980	0.984	0.960	n/a	n/a	n/a
PersFL	0.962	0.974	0.948	0.960	0.962	0.962

final-round AUC shifts by at most 0.012 on the MLP and 0.003 on logistic regression, slightly upward if anything and inside seed-to-seed variance, so for rank-ordering the noise is essentially free. Sparsification costs 0.02 to 0.05 AUC on the MLP and around 0.005 on LR. The worst case is FedNova with sparsification, which reaches a final F1 of only 0.009 and is unusable regardless of the AUC curve. At the top end the MLP reaches 0.984 AUC (FedNova with noise) against 0.957 for LR; closing that gap costs 10 to 25 times more compute per round, so when communication is the constraint LR with sparsification is reasonable.

These are robustness numbers; privacy is not claimed. There is no Rényi-DP accountant, no subsampling amplification argument and no auditable ε at the end of training, so Table V labels the mechanism Gaussian noise. Turning these figures into a guarantee needs an accountant.

E. From Rankings to Decisions: the Triage Layer (RQ5)

The two-convention result leaves a practical question: if no deployable strategy reliably beats the field, what should a fraud team run? Our answer optimises the decision instead of the ranking. The triage layer applies, per bank, a split-conformal clear threshold at a recall-80 target (with a plain-threshold fallback where calibration frauds are too few), a Chow-style defer band [28] whose width follows $\rho = c_{FN}/c_{FP}$, and a 5% review-capacity cap. Everything below the band auto-clears, the band defers to human review, and the rest alerts.

Table VI pools 3 $\alpha \times 5$ seeds $\times$ 6 ρ per dataset. Three properties matter. First, the auto-clear rate is the dataset’s difficulty made visible: 99.8% of ULB transactions clear at recall 80, 89% on IBM, 74% on SAML-D. Second, the defer rate barely moves (3.8 to 4.6%): the capacity cap binds. Third, expected cost per transaction moves by only 0.9 to 3.6% while ρ sweeps from 1 to 50, so a team that cannot pin its cost ratio within a factor of fifty still lands on the same policy. Conformal validity holds for 64 to 79% of bank-cells, the rest falling back to threshold mode; the soft spot is ULB $\alpha = 0.5$, where the test window holds only 52 frauds.

TABLE VI
TRIAGE LAYER AT RECALL-80, POOLED OVER α , SEEDS AND ρ

	ULB	IBM HI-S	SAML-D
Auto-clear rate	0.998	0.894	0.735
Defer rate	0.038	0.045	0.046
Policy miss rate	0.081	0.050	0.074
Conformal coverage valid	0.64	0.79	0.75
Cost drift, $\rho: 1 \rightarrow 50$	+3.6%	+1.5%	+0.9%

F. Per-Bank Fairness

Across every populated ULB noise-grid cell, Jain’s index sits at 0.9999 or better and σ -AUC stays under 0.02, so the aggregation itself treats banks evenly. The sweep-level picture is harsher: at $\alpha \leq 0.10$, at least one bank posts a worst-case F1 of zero in 110/110 ULB cells, 104/110 SAML-D cells and 83/110 IBM cells (36, 19 and 3 of 55 at $\alpha = 0.5$). These zeroes are a property of the data allocation, and they make the strongest case we know for per-bank triage thresholds.

IV. DISCUSSION

A. Principal Findings

RQ1’s answer is that no single degradation curve exists; the shape belongs to the dataset. On ULB all eleven strategies decline smoothly in lockstep as α falls; on SAML-D the decline is stochastic, driven by the partition draw; on IBM HI-Small heterogeneity barely registers. Verifying partition structure therefore matters more than choosing among FedAvg-family aggregators.

The fusion result must be read under two conventions: the best deployable Friedman rank exceeds the FL family mean in every cell, yet never the per-cell best backbone, whose identity shifts with the seed. The gates are weighted ensembles whose value is hedging against the unanswerable question of which backbone a given quarter’s data will favour. PersFL, meanwhile, buys a performance floor at the cost of the single global model that is FL’s principal communication advantage.

The cost and triage results prove the most broadly applicable. Best-AUPRC and cost-best coincide in only 10 of 81 cells, so a leaderboard alone misallocates review budget in the other 71. None of this depended on a superior model, only a better decision rule.

B. Limitations

(1) The noise ablation is a robustness experiment; no differential-privacy guarantee is claimed anywhere in this paper, and the grid is single-seed and predates the canonical pipeline. (2) The Typology-Aware gate uses ground-truth fraud-type labels at inference and is an upper bound only. (3) The centralised baselines are single pooled runs, and the centralised LightGBM path is degenerate and excluded pending a fix. (4) Five seeds keeps every per- α Wilcoxon at or above $p = 0.0625$ by construction; the pooled tests carry the inference, and per-cell standard deviations remain wide. (5) Conformal validity holds for 64 to 79% of bank-cells, with ULB $\alpha = 0.5$ the weakest slice. (6) At $\alpha = 0.05$ the Dirichlet sampler legitimately allocates zero fraud

cases to some banks, flooring per-bank evaluation. (7) The banks are simulated silos and the heterogeneity is label-prevalence skew, whereas real silos also differ in features and schemas, so “cross-bank” and “deployable” are within-simulation claims and a pilot on real institutions remains future work.

C. Comparison with Prior Work

Across the 42 papers retained under the PRISMA protocol [8], most report AUPRC on a single dataset with a single seed against one or two FL baselines; oracle and deployable gating are rarely distinguished, and asymmetric loss rarely considered. This benchmark expands along all of those dimensions, and its two-convention reporting separates “beats the average” from “beats the best”.

V. CONCLUSION AND FUTURE WORK

We asked whether FL with an expert-fusion layer on top is an efficient architecture for cross-bank fraud detection. Four findings emerge. First, the benchmark: 516 runs over three dataset families, eleven strategies, three Dirichlet regimes and five seeds, with partitions reproducible cell by cell. Second, a two-convention verdict on the gate: best deployable Friedman rank and a win over the FL family mean on all three datasets, no win over the per-cell best backbone, near-uniform weights; the gate is a computationally inexpensive hedge against backbone selection. Third, a survival map rewritten as frequencies: collapse at extreme skew is a partition-induced instability, FedNova’s operating point is the most fragile, and PersFL never hits $F1 = 0$. Fourth, a triage layer that turns a flat ranking into a policy, auto-clearing 74 to 99.8% of transactions at recall 80 with a 4 to 5% defer rate.

Several directions follow from the limitations. The most immediate is adding a real differential-privacy mechanism with an accountant so the gradient-noise robustness numbers can become privacy numbers. Three extensions would deepen the benchmark: a genuinely per-sample learned gate, the only variant that showed any signal in our post-hoc probes; a federated gradient-boosting baseline to attack the residual tree advantage on the AML datasets; and a canonical-protocol rerun at IBM’s larger scales. Finally, scaling past four banks and piloting under Australian data-locality rules would test these findings at deployment scale.

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REFERENCES

- [1] B. Unger and J. Ferwerda, *Money Laundering in the Real Estate Sector*. Cheltenham, U.K.: Edward Elgar, 2011.
- [2] B. Oztas et al., “Enhancing anti-money laundering: Development of a synthetic transaction monitoring dataset (SAML-D),” in *Proc. IEEE ICEBE*, 2023, pp. 47–54.
- [3] P. Reuter and E. M. Truman, *Chasing Dirty Money: The Fight Against Money Laundering*. Washington, DC, USA: Peterson Inst. Int. Econ., 2004.
- [4] Australian Prudential Regulation Authority, “Prudential standard CPS 234: Information security,” APRA, Sydney, Australia, 2019.
- [5] H. B. McMahan et al., “Communication-efficient learning of deep networks from decentralized data,” in *Proc. AISTATS*, vol. 54, 2017, pp. 1273–1282.
- [6] Y. Zhao et al., “Federated learning with non-IID data,” 2018, arXiv:1806.00582.
- [7] S. P. Karimireddy et al., “SCAFFOLD: Stochastic controlled averaging for federated learning,” in *Proc. ICML*, vol. 119, 2020, pp. 5132–5143.
- [8] M. J. Page et al., “The PRISMA 2020 statement: An updated guideline for reporting systematic reviews,” *BMJ*, vol. 372, Art. no. n71, 2021.
- [9] T. Li et al., “Federated optimization in heterogeneous networks,” in *Proc. MLSys*, vol. 2, 2020, pp. 429–450.
- [10] J. Wang et al., “Tackling the objective inconsistency problem in heterogeneous federated optimization,” in *NeurIPS*, vol. 33, 2020, pp. 7611–7623.
- [11] A. Fallah, A. Mokhtari, and A. Ozdaglar, “Personalized federated learning with theoretical guarantees: A model-agnostic meta-learning approach,” in *NeurIPS*, vol. 33, 2020, pp. 3557–3568.
- [12] T. Li et al., “Ditto: Fair and robust federated learning through personalization,” in *Proc. ICML*, 2021, pp. 6357–6368.
- [13] T.-M. H. Hsu, H. Qi, and M. Brown, “Measuring the effects of non-identical data distribution for federated visual classification,” 2019, arXiv:1909.06335.
- [14] R. A. Jacobs, M. I. Jordan, S. J. Nowlan, and G. E. Hinton, “Adaptive mixtures of local experts,” *Neural Comput.*, vol. 3, no. 1, pp. 79–87, 1991.
- [15] N. Shazeer et al., “Outrageously large neural networks: The sparsely-gated mixture-of-experts layer,” in *Proc. ICLR*, 2017.
- [16] Y. Zhou et al., “Mixture-of-experts with expert choice routing,” in *NeurIPS*, vol. 35, 2022, pp. 7103–7114.
- [17] T. Chen and C. Guestrin, “XGBoost: A scalable tree boosting system,” in *Proc. ACM SIGKDD*, 2016, pp. 785–794.
- [18] G. Ke et al., “LightGBM: A highly efficient gradient boosting decision tree,” in *NeurIPS*, vol. 30, 2017, pp. 3146–3154.
- [19] L. Prokhorenkova et al., “CatBoost: Unbiased boosting with categorical features,” in *NeurIPS*, vol. 31, 2018, pp. 6638–6648.
- [20] A. Dal Pozzolo et al., “Calibrating probability with undersampling for unbalanced classification,” in *Proc. IEEE SSCI*, 2015, pp. 159–166.
- [21] E. Altman et al., “Realistic synthetic financial transactions for anti-money laundering models,” in *NeurIPS Datasets and Benchmarks*, vol. 36, 2023.
- [22] N. V. Chawla et al., “SMOTE: Synthetic minority over-sampling technique,” *J. Artif. Intell. Res.*, vol. 16, pp. 321–357, 2002.
- [23] M. Abadi et al., “Deep learning with differential privacy,” in *Proc. ACM CCS*, 2016, pp. 308–318.
- [24] R. C. Geyer, T. Klein, and M. Nabi, “Differentially private federated learning: A client-level perspective,” 2017, arXiv:1712.07557.
- [25] Y. Lin et al., “Deep gradient compression: Reducing the communication bandwidth for distributed training,” in *Proc. ICLR*, 2018.
- [26] J. Cohen, *Statistical Power Analysis for the Behavioral Sciences*, 2nd ed. Hillsdale, NJ, USA: Lawrence Erlbaum, 1988.
- [27] J. Demšar, “Statistical comparisons of classifiers over multiple data sets,” *J. Mach. Learn. Res.*, vol. 7, pp. 1–30, 2006.
- [28] C. K. Chow, “On optimum recognition error and reject tradeoff,” *IEEE Trans. Inf. Theory*, vol. 16, no. 1, pp. 41–46, 1970.
- [29] V. Vovk, A. Gammerman, and G. Shafer, *Algorithmic Learning in a Random World*. New York, NY, USA: Springer, 2005.
- [30] T. Yoon, S. Shin, S. J. Hwang, and E. Yang, “FedMix: Approximation of mixup under mean augmented federated learning,” in *Proc. ICLR*, 2021.
- [31] O. Marfoq et al., “Federated multi-task learning under a mixture of distributions,” in *NeurIPS*, vol. 34, 2021, pp. 15434–15447.
- [32] C. Dun et al., “FedJETs: Efficient just-in-time personalization with federated mixture of experts,” 2023, arXiv:2306.08586.
- [33] D. Chen, D. Gao, W. Kuang, Y. Li, and B. Ding, “pFL-Bench: A comprehensive benchmark for personalized federated learning,” in *NeurIPS Datasets and Benchmarks*, vol. 35, 2022.
- [34] K. Cheng et al., “SecureBoost: A lossless federated learning framework,” *IEEE Intell. Syst.*, vol. 36, no. 6, pp. 87–98, 2021.
- [35] W. Yang et al., “FFD: A federated learning based method for credit card fraud detection,” in *Proc. Int. Congr. Big Data*, 2019, pp. 18–32.